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Applied Soft Computing

Manuscript title:

Battlefield-Adaptive Evolutionary Algorithm with Online Reference-Reallocation and Quantile ε -Grid Selection for Large-Scale and Constrained Multi-Objective Optimization

Manuscript ID: [to be assigned on submission]

Date: [insert submission date]

Dear Editor,

On behalf of the author, I am pleased to submit the enclosed manuscript entitled “*Battlefield-Adaptive Evolutionary Algorithm with Online Reference-Reallocation and Quantile ε -Grid Selection for Large-Scale and Constrained Multi-Objective Optimization*” for consideration for publication in *Applied Soft Computing*.

This paper proposes VOR, a dimension-adaptive evolutionary algorithm whose central contribution is a *battlefield routing* rule that services the large-scale ($D = 300$) and the constrained low-dimensional ($D = 10\text{--}60$) battlefields from a *single codebase*. On the six-problem benchmark with the four traditional baselines and our EDD lineage, VOR attains the first IGD rank on three of six problems at 30 seeds, with a statistically significant advantage over EDD on ZDT1, LSMOP1 and CF1 (30-seed Wilcoxon, $p < 10^{-5}$). Very importantly, we position the result honestly: against the 2026 SOTA trio (FDSEA/GDVTSE/MOEA-IB, the same 30-seed protocol), FDSEA outperforms VOR on all three $D = 300$ instances; VOR’s differentiating value is the *single-codebase coverage* of the constrained low-dimensional battlefield where the three SOTA algorithms have no published $D = 10\text{--}60$ benchmark.

We believe this manuscript fits *Applied Soft Computing* well for three reasons. First, the contribution is a *practical* one: a single codebase that removes the need to re-implement a dimension adaptive algorithm for a new problem class. Second, the experimental protocol is statistically sound (30 seeds, Friedman and Wilcoxon significance tests, 5-algorithm + 3-SOTA comparison set). Third, the honest-boundary reporting (the EDD-family seed-invariance on MAF14, the DTLZ2-300D EDD margin, the CF1 decomposition structural advantage) matches the journal’s emphasis on reproducible, well-characterized application research rather than a single claimed champion algorithm.

The manuscript has not been published and is not under consideration elsewhere. The corresponding author is Yuxuan Zhang (3150644070@qq.com). The implementations, the full benchmark suite, the complete 900-file 30-seed result archive (6 problems $\times$ 5 algorithms $\times$ 30 seeds), all statistical outputs, and the publication-grade figures are publicly available at <https://github.com/qingdi-miruisi/VOR-battlefield-ea> (public GitHub repository, MIT license; one-command reproduction via the README.md).

We would be happy to suggest potential reviewers. Our suggested reviewers (no conflicts with the author) are:

Name 1 , [Institution 1] — [email 1]

Name 2 , [Institution 2] — [email 2]

Name 3 , [Institution 3] — [email 3]

Thank you for considering this manuscript.

Sincerely, Yuxuan Zhang

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